Challenge Details and Requirements

Challenge Overview: OBV Bullish Divergence

In this exercise, we use the **On-Balance Volume (OBV) Bullish Divergence** event as the target factor. You will work with the mocked data to implement the event generator logic and the performance evaluation process, then provide the event factor performance analysis.

Concept

On-Balance Volume (OBV) is defined as cumulative volume based on close-price direction:

- If today's close price > yesterday's close price: add today's volume to OBV
- If today's close price < yesterday's close price: subtract today's volume from OBV
- If unchanged: OBV stays the same

An **OBV Bullish Divergence** event (bullish reversal alert) is triggered when both conditions are met for a rolling window:

- **Price trend** is negative
- **OBV trend** is positive
- **Rolling Window** Size: 7 trading days

Reference: Investopedia — On Balance Volume

Mocked Dataset

All data needed for the task are located in `data/` folder:

File	Description
`daily_prices.csv`	8 stocks (`osid` 1001-1008), 120 trading days (2024-01-02 to 2024-06-17). Fields: `osid` , `trade_date` , `priceopen` , `pricehigh` , `pricelow` , `priceclose` , `volume`
`benchmark_returns.csv`	Daily benchmark return (`bench_return`) and benchmark close (`bench_close`) for alpha calculation

File	Description
`stock_metadata.csv`	`ticker`, `sector`, `sector_group`, `style_group` for optional grouped analysis

Deliverables

Submit a small Python project (script or notebook) that completes the tasks below, plus a short **`RESEARCH_NOTE.md`** (<= 2 pages) for the event performance analysis + research question answering for Part C questions.

Part A — Implement the Event Generator (about 35 minutes)

Write `solution.py` (or an equivalent file) and complete the following:

1. Read `data/daily_prices.csv`
2. Compute OBV by `osid`
3. Compute 7-day rolling trend signs for both price and OBV (better use linear regression slope sign)
4. Identify OBV bullish divergence signals and write **`output/events.csv`** with fields: `event_id`, `event_date`, `osid`

Part B — Validate Event Performance (about 35 minutes)

Extend your code to evaluate the event performance, generate **`output/event_returns.csv`** and **`output/performance_summary.json`**.

Please use the **5th-trading-day** performance as the evaluation outcome, basic metrics should include but not limited by:

1. **event's 5th forward day return**
2. **event's 5th forward day benchmark-relative alpha**
3. **event's 5th forward day** event effectiveness

Note: If an `osid` has fewer than 5 trading days after the event date, skip that event.

Optional bonus: Additional performance analysis on specific sector_groups.

Part C — Research Discussion (about 20 minutes)

In `RESEARCH_NOTE.md`, answer **all three** questions below. Keep answers tied to *this specific factor* where possible. No code implementation is required for this part.

1. **Time-Series vs. Cross-Sectional Data**
   This event study uses panel data (multiple stocks x multiple dates). What additional considerations are needed versus pure cross-sectional modeling? How did you prevent look-ahead bias in Part A and Part B?

2. **Overfitting / Validation**
   Suppose this bullish-divergence factor performs well in-sample but weakly out-of-sample. What steps would you take before recommending it to a portfolio manager?

3. **Simple Models vs. Complex Models**
   Would you replace this rule-based OBV divergence signal with an LSTM or gradient boosting model? Under what conditions would you replace it, and under what conditions would you keep the interpretable rule?

Recommended Deliverables Structure

```bash
challenge/
  output/
    events.csv
    event_returns.csv
    performance_summary.json
  solution.py          # or solution.ipynb
  RESEARCH_NOTE.md
  requirements.txt     # include any additional dependencies
```

How to Run

```bash
cd challenge
python -m venv .venv
# Windows: .venv\\Scripts\\activate
# macOS/Linux: source .venv/bin/activate
```

```
pip install -r requirements.txt
python solution.py
```

Evaluation Criteria

Dimension	What We Evaluate
Correctness	Event logic matches specification; output is reproducible
Point-in-time discipline	No future data leakage in signals or labels
Code quality	Clear structure; appropriate pandas usage
Performance analysis	Correct forward return, alpha, and other statistics analysis
Investment judgment	Research note shows quantitative finance maturity
AI tool usage	Effective use of tools with independent validation